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spring boot

Java & Spring– Case Study Based Project Execution

Instructor led Training

What is this course about?

This training provides a **hands-on, case study–based learning experience on building a Digital Banking Microservice using Java and Spring technologies**. Participants will design and implement backend banking services such as account management, transaction processing, and balance enquiry using **Spring Boot, REST APIs, and database integration**. The program follows a **learning-by-doing approach**, where participants build a working banking service while learning modern Java development practices.

Key Features

1. **Duration:**

   Live-online/Classroom ILT: 10 Hours
2. **Level:** Intermediate
3. **Language:** English
4. **Prerequisites:**
   - Basic knowledge of Java programming
   - Understanding of object-oriented programming concepts
   - Familiarity with web application concepts and REST APIs
   - Basic knowledge of databases and SQL
   - Prior exposure to Spring Framework or Spring Boot (preferred but not

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mandatory)

What are the objectives of Java & Spring– Case Study Based Project Execution?

At the end of this training, you will be able to:

- Understand modern banking application architecture and microservices design
- Develop backend services using Java, Spring Framework, and Spring Boot
- Design and implement RESTful APIs for banking services
- Build modules for account management, transactions, and balance enquiry
- Integrate applications with SQL/PostgreSQL/MongoDB databases
- Implement API validation, exception handling, and service integration
- Test APIs and validate application workflows using Postman

Who is this Training for?

- Java Developers
- Backend Developers
- Full Stack Developers working with Java technologies
- Software Engineers building microservice-based applications
- Professionals involved in API development and backend system design

Course Outline:

Program Introduction and Context

Modern banking applications rely on scalable microservices, API-driven architectures, and datadriven systems to support digital banking experiences such as account management, payments, loan processing, and transaction monitoring.

Java-based technologies such as Spring Framework, Spring MVC, Spring Boot, REST APIs, and modern databases (SQL/PostgreSQL/MongoDB) are widely used to build secure, scalable, and reliable banking systems.

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This program is designed to provide participants with a hands-on, project-driven learning experience where they will build a banking application service using the modern Java technology stack. The training will follow a learning-by-doing model, where participants execute 90% of the implementation while the trainer facilitates, guides, and reviews the architecture and coding practices.

Participants will work on a banking domain case study, implementing real-world features such as account services, transaction management, and API-based banking services. The focus is on practical application development, API design, database integration, and secure service implementation.

Banking Domain Case Study Project

Project Title - Digital Banking Microservice – Account & Transaction Service

Project Overview

Participants will build a backend service for a digital banking platform that supports:

- Customer account management
- Transaction processing
- Balance enquiry
- REST-based service integration
- Database persistence using SQL/PostgreSQL/MongoDB

The project simulates a banking service used by mobile or internet banking applications.

Project Functional Modules

1. Customer Account Management
2. Transaction Processing
3. Balance Enquiry Service
4. Transaction History
5. REST API Integration
6. Database persistence

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Technology Stack

- Java
- Spring Framework
- Spring MVC
- Spring Boot
- REST API Development
- SQL / PostgreSQL / MongoDB

Session Plan (10 Hours)

Session 1 – Banking System Architecture & Project Setup
Duration: 2 Hours

- Banking application architecture overview
- Microservice architecture in banking systems
- Spring Boot project structure
- Dependency management (Maven / Gradle)
- Project setup and configuration

Project Activity

- Create a Spring Boot project
- Configure dependencies
- Setup project structure for banking services

Session 2 – Customer Account Management Service Duration: 2 Hours
Topics

- Spring MVC architecture
- Controller and service layers
- Entity modeling for banking accounts
- Database connectivity

Project Activity

- Create Account entity model
- Implement Create Account API
- Connect application to SQL/PostgreSQL/MongoDB

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- Store account information in database

Session 3 – Transaction Processing APIs Duration: 2 Hours

- REST API design principles
- Transaction service design
- Service layer implementation
- API validation

Project Activity

- Implement Deposit API
- Implement Withdraw API
- Validate transaction requests
- Update account balances

Session 4 – Balance & Transaction History Services Duration: 2 Hours

- RESTful service integration
- Querying database for transactions
- Exception handling in Spring Boot
- API testing

Project Activity

- Implement Balance enquiry API
- Implement Transaction history API
- Test APIs using Postman

Session 5 – Final Integration & Case Study Execution Duration: 2 Hours

- Service integration and validation
- API security considerations
- Performance and reliability considerations

Project Activity

- Integrate all services
- Test the banking workflow end-to-end
- Demonstrate the working application

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